

7.1 Networking Over TCP/IP

There has been a wide-scale implementation of the so-called broadband setup in India. Many users find broadband service affordable, and have switched from dial-up to Ethernet-based “broadband” connections. This kind of setup involves a LAN (Local Area Network) connection that is routed to a local gateway, which acts as your proxy to the Internet. Here, you need to know what IP addresses, subnet masks, gateways and DNS or nameservers are.

7.1.1 Fundamentals of TCP/IP

Networks are essentially of two types: public and private networks. The latter is used in local networks such as homes, offices and cyber cafes. This class of IP cannot be directly used to connect to the Internet. This is where your ISP—the Internet Service Provider—comes in. Their Domain Name Server (DNS) assigns a public IP to your machine, using which you surf the Internet. Public IPs have to be registered with the Network Information Centre (NIC) to avoid IP address conflicts. An IP is your address on the Internet, using which data packets can be sent to and from your computer.

A private IP is accessible only within a LAN. For this purpose, the NIC has reserved certain address ranges that will never be registered publicly. In short, there can be more than one identical private IP in different networks, but only within LANs. The Web sites that you regularly visit are actually public IPs. For example, the name `www.yahoo.com` is resolved to an IP address; this address is public—it is accessible by anyone in the world. The names are assigned for the sake of simplicity because it's not practical to remember the address of such sites.

Every IP address is represented as a “dotted quad” (in case of IPv4) as `x.x.x.x`. This can be split into two parts: the network portion and the host portion. The network portion is the same for all nodes (computers or printers) connected to a LAN. The host portion distinguishes each node from the other. The subnet mask